



US012412208B2

(12) **United States Patent**
van Horne et al.

(10) **Patent No.:** **US 12,412,208 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **OPTICAL SCANNING FOR WEIGHTS AND MEASURES**

(71) Applicant: **Maplebear Inc.**, San Francisco, CA
(US)

(72) Inventors: **Camille van Horne**, San Francisco, CA
(US); **Sean Cashin**, San Francisco, CA
(US)

(73) Assignee: **Maplebear Inc.**, San Francisco, CA
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/213,474**

(22) Filed: **Jun. 23, 2023**

(65) **Prior Publication Data**

US 2023/0334556 A1 Oct. 19, 2023

Related U.S. Application Data

(63) Continuation of application No. 17/068,594, filed on
Oct. 12, 2020, now Pat. No. 11,727,470.

(51) **Int. Cl.**

G06Q 30/0601 (2023.01)
G06F 18/214 (2023.01)
G06Q 10/0875 (2023.01)
G06Q 20/04 (2012.01)
G06Q 20/08 (2012.01)

(Continued)

(52) **U.S. Cl.**

CPC **G06Q 30/0635** (2013.01); **G06F 18/214**
(2023.01); **G06Q 10/0875** (2013.01); **G06Q**
20/047 (2020.05); **G06Q 20/085** (2013.01);
G06Q 30/0206 (2013.01); **G06Q 30/0641**
(2013.01); **G06Q 40/12** (2013.12); **G06V**
10/993 (2022.01); **G06V 30/416** (2022.01);

(Continued)

(58) **Field of Classification Search**

CPC G06Q 20/322; G06Q 10/10; G06Q 30/08;
G06V 30/413; G06V 30/19013; G06V
10/40; G06F 40/295; G06F 16/5838
USPC 1/1; 705/26.8; 235/383; 382/209
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

11,257,049 B1 * 2/2022 Durazo Almeida . G06V 30/413
2003/0102373 A1 * 6/2003 Swartz G07G 1/0054
235/383

(Continued)

OTHER PUBLICATIONS

Canadian Intellectual Property Office, Office Action, Canadian
Patent Application No. 3,188, 103, Jun. 18, 2024, 5 pages.

(Continued)

Primary Examiner — Olusegun Goyea

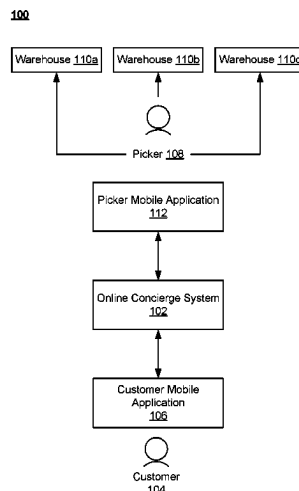
(74) *Attorney, Agent, or Firm* — Fenwick & West LLP

(57)

ABSTRACT

An online concierge system sends an order associated with a customer to a shopper for fulfillment at a store. The order specifies an ordered amount of an item. The online concierge system receives an image of a receipt for the order from the shopper after fulfillment of the order, applies an image processing algorithm to identify the item in the image of the receipt, and identifies a measured quantity within the image. The measured quantity represents an actual amount of the item purchased at the store. The online concierge system determines a difference between the actual amount and the ordered amount of the item, and determines an amount to be charged or reimbursed to the customer based in part on the difference between the actual amount and the ordered amount of the item.

20 Claims, 14 Drawing Sheets



(51) **Int. Cl.**

G06Q 30/0201 (2023.01)
G06Q 40/12 (2023.01)
G06V 10/98 (2022.01)
G06V 30/416 (2022.01)
G06Q 10/083 (2023.01)
G06Q 20/32 (2012.01)
G06V 30/10 (2022.01)

(52) **U.S. Cl.**

CPC *G06Q 10/083* (2013.01); *G06Q 20/326*
(2020.05); *G06V 30/10* (2022.01)

(56) **References Cited**

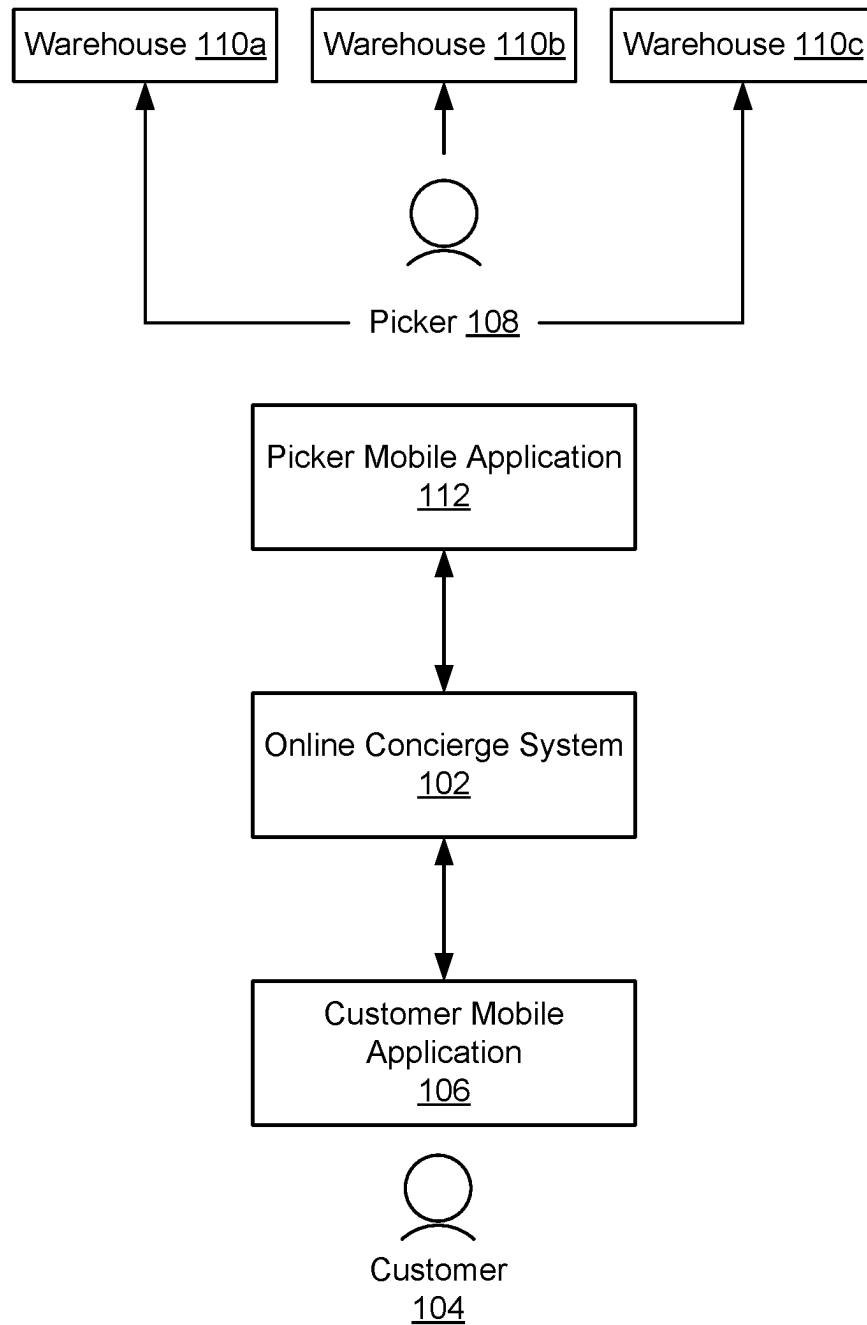
U.S. PATENT DOCUMENTS

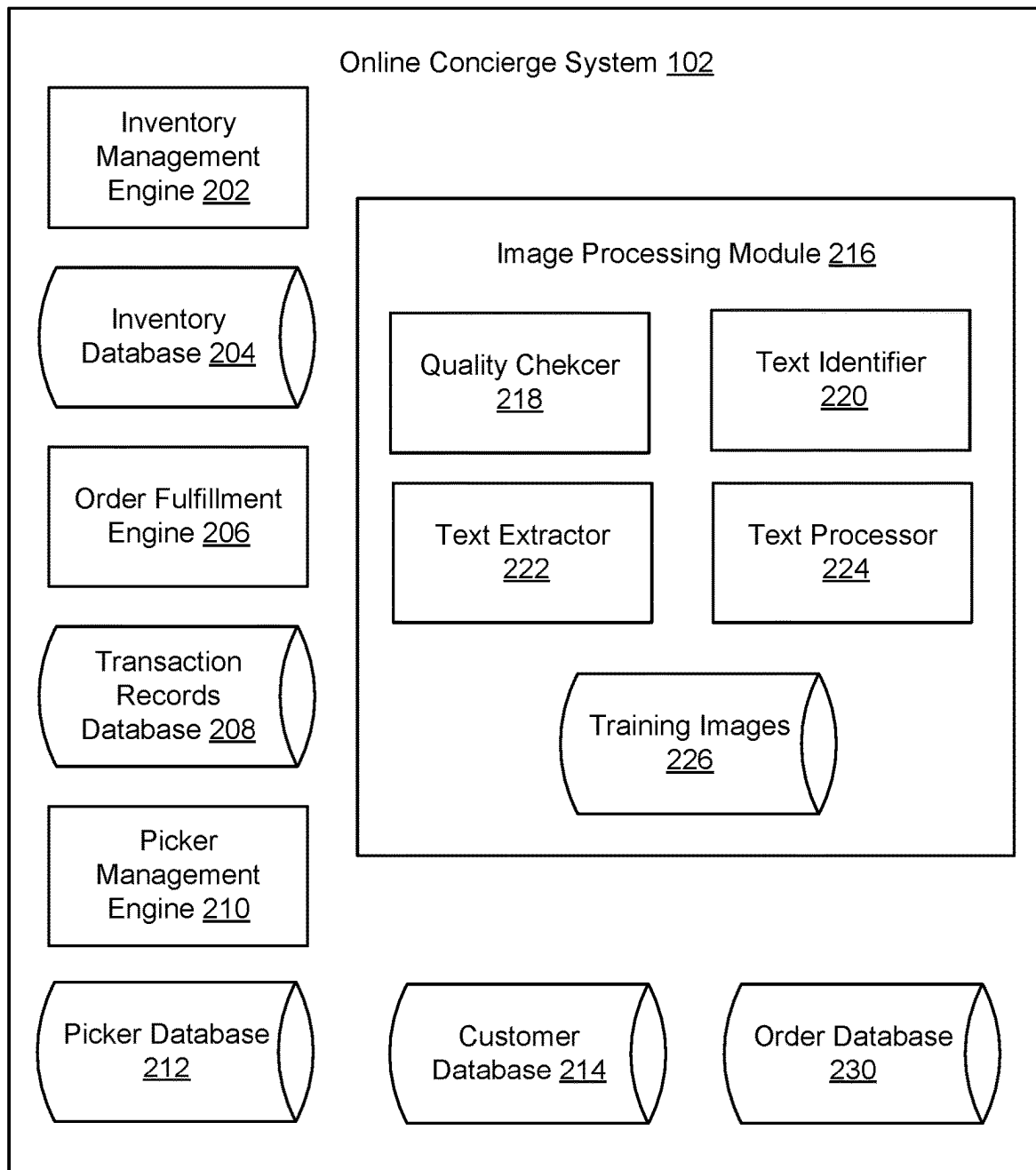
2008/0152231 A1 * 6/2008 Gokturk G06Q 30/08
382/209
2008/0265026 A1 10/2008 Silverbrook et al.
2013/0080289 A1 * 3/2013 Roy G06Q 20/322
705/26.8
2016/0086255 A1 3/2016 Sainfort et al.
2018/0189786 A1 7/2018 Poole
2020/0057903 A1 * 2/2020 Leizerowich H04N 23/80
2021/0158038 A1 * 5/2021 Shanmuganathan .. G06V 30/20

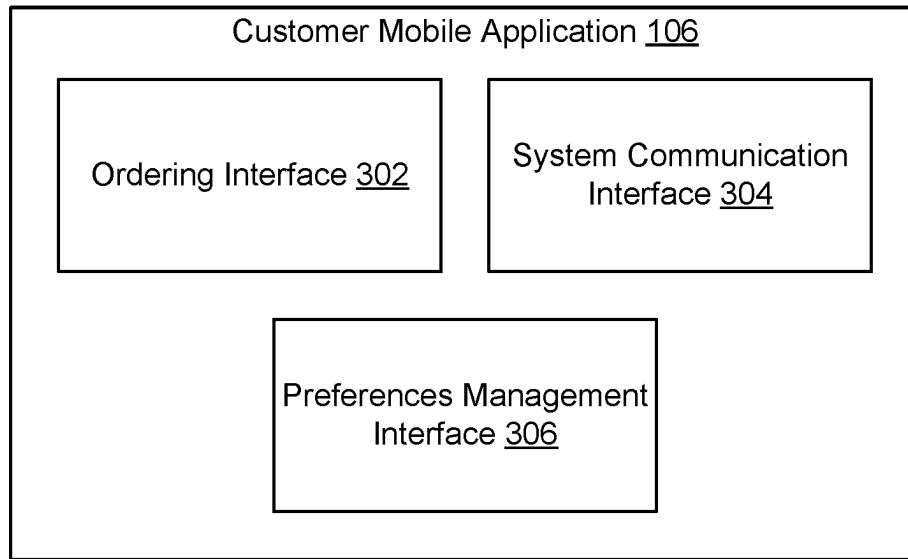
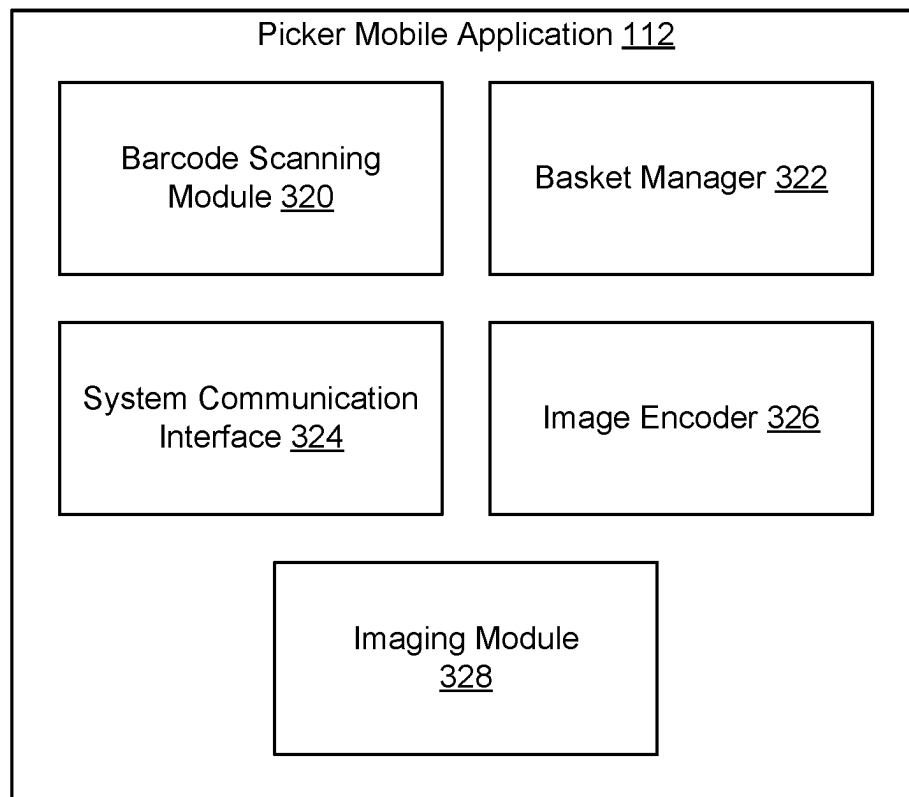
OTHER PUBLICATIONS

PCT International Search Report and Written Opinion, PCT Application No. PCT/US2021/054418, Jan. 21, 2022, 13 pages.
Office Action, U.S. Appl. No. 17/068,594, filed Sep. 16, 2022, 8 pages.

* cited by examiner

100**FIG. 1**

**FIG. 2**

**FIG. 3A****FIG. 3B**

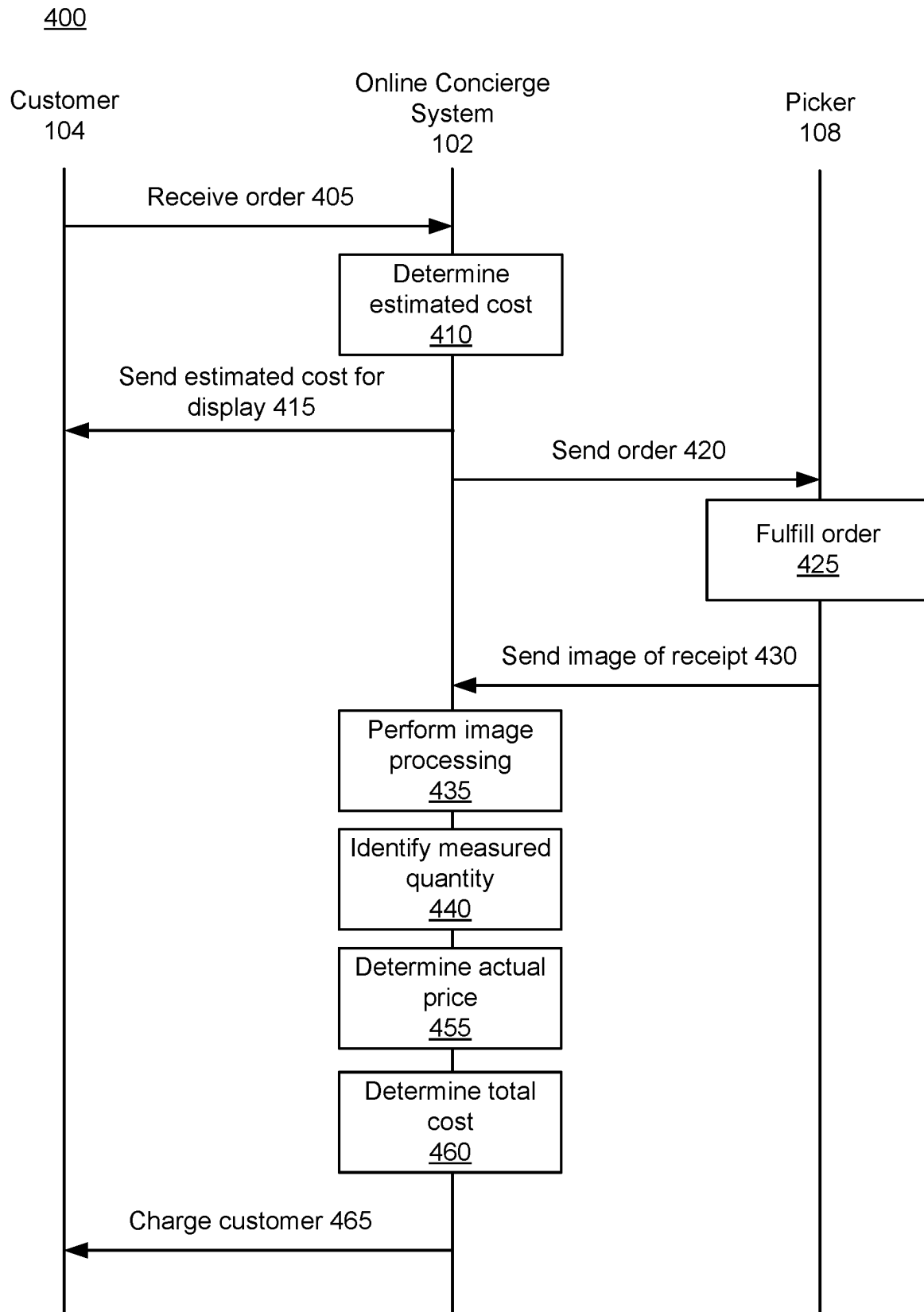


FIG. 4

500

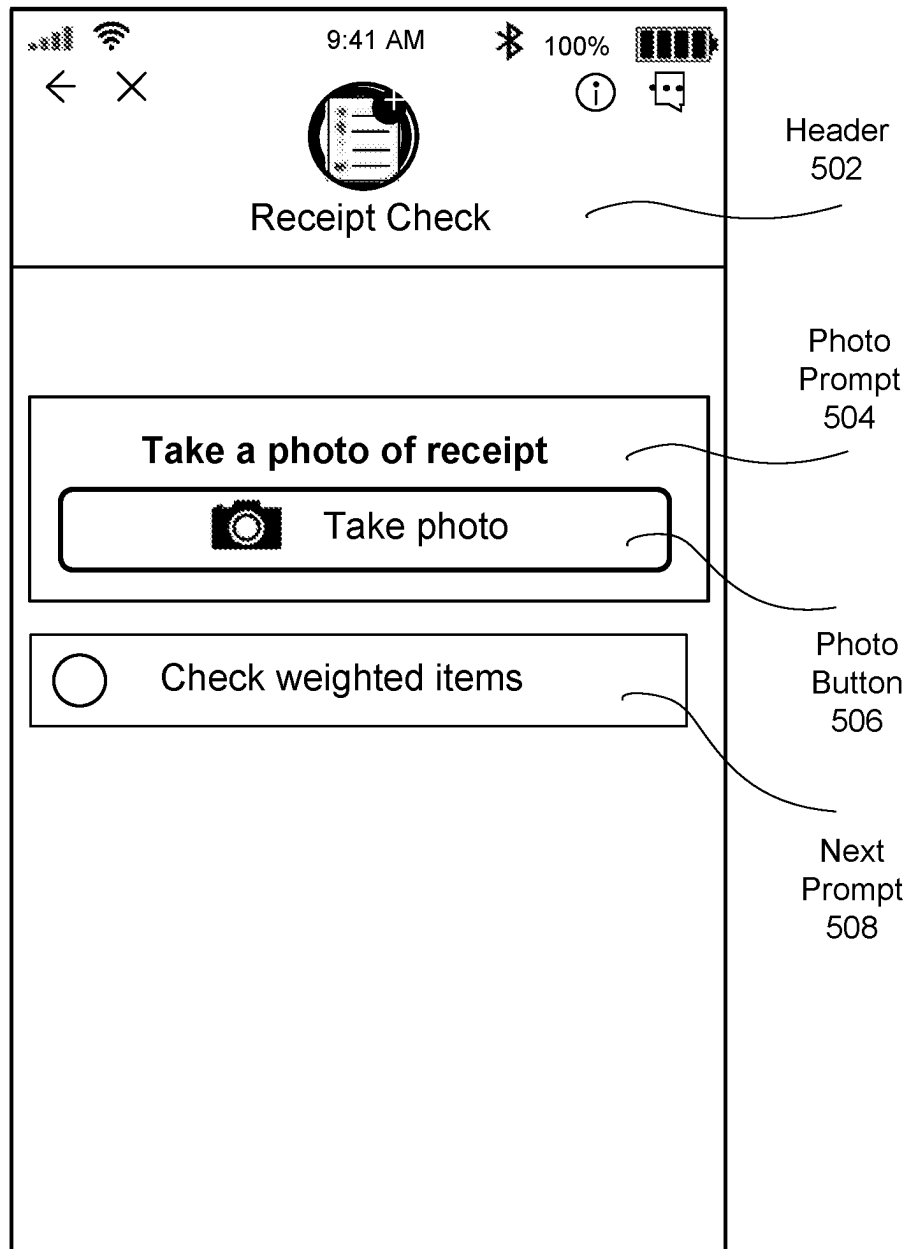


FIG. 5A

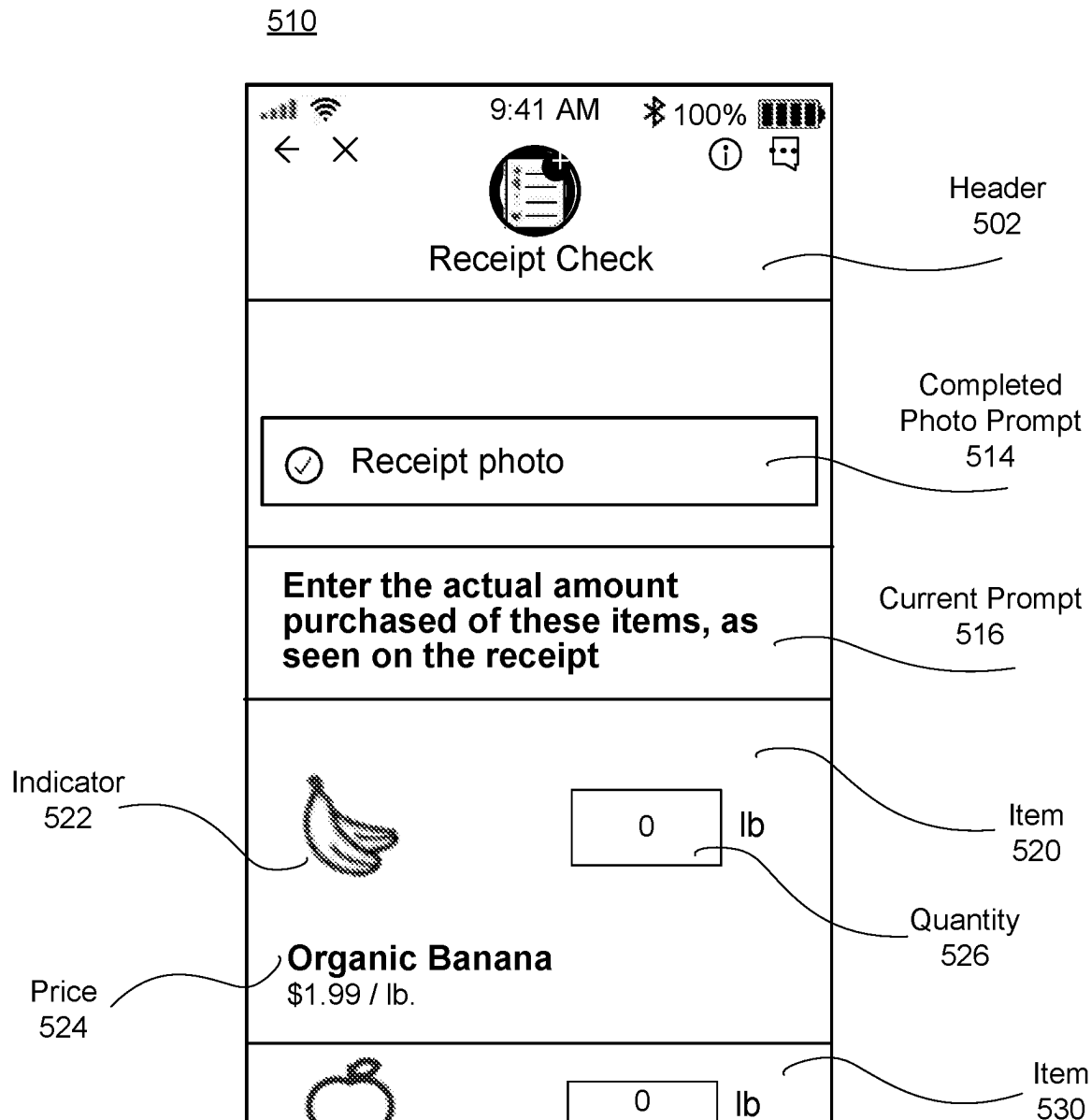


FIG. 5B

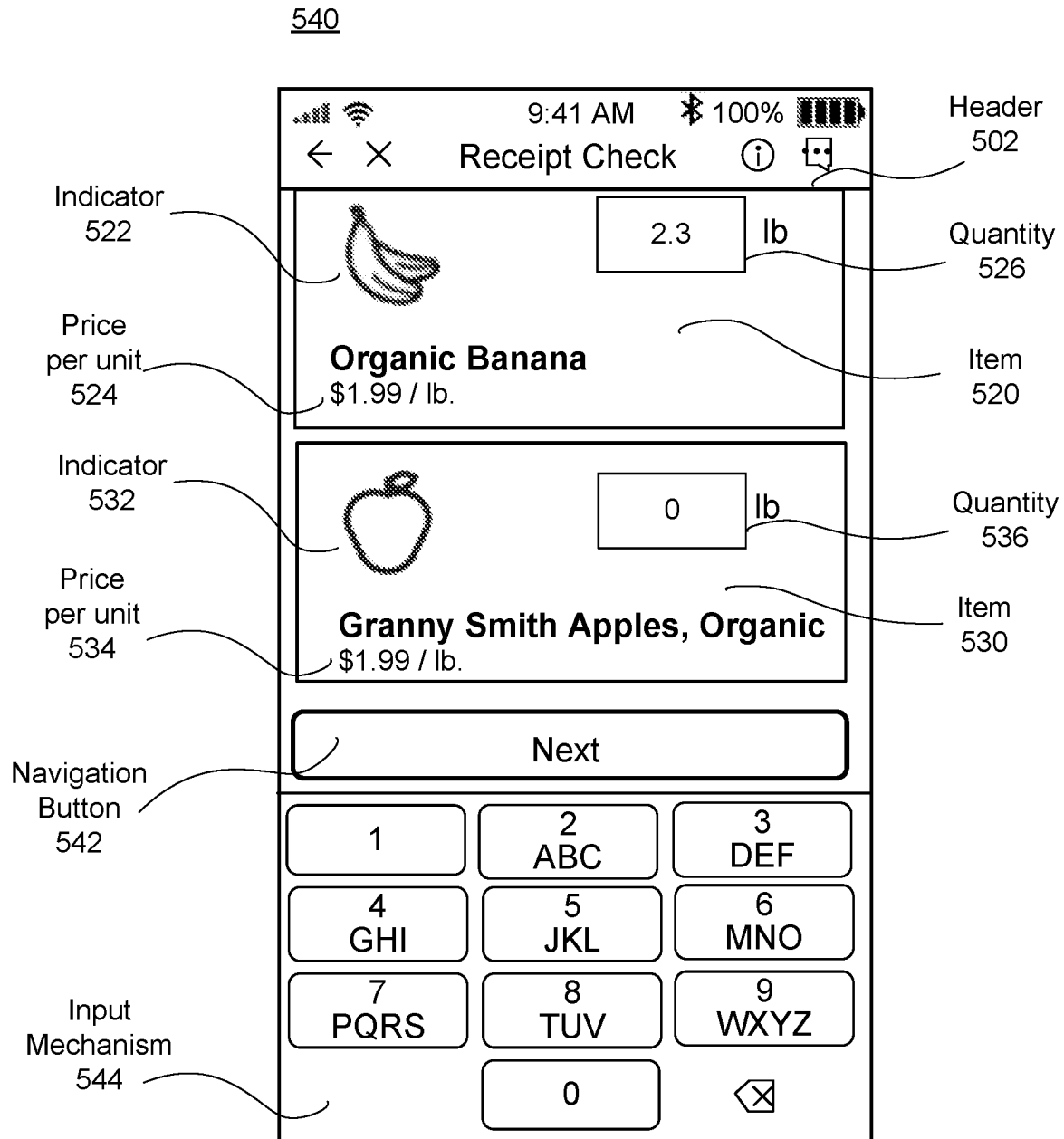


FIG. 5C

600

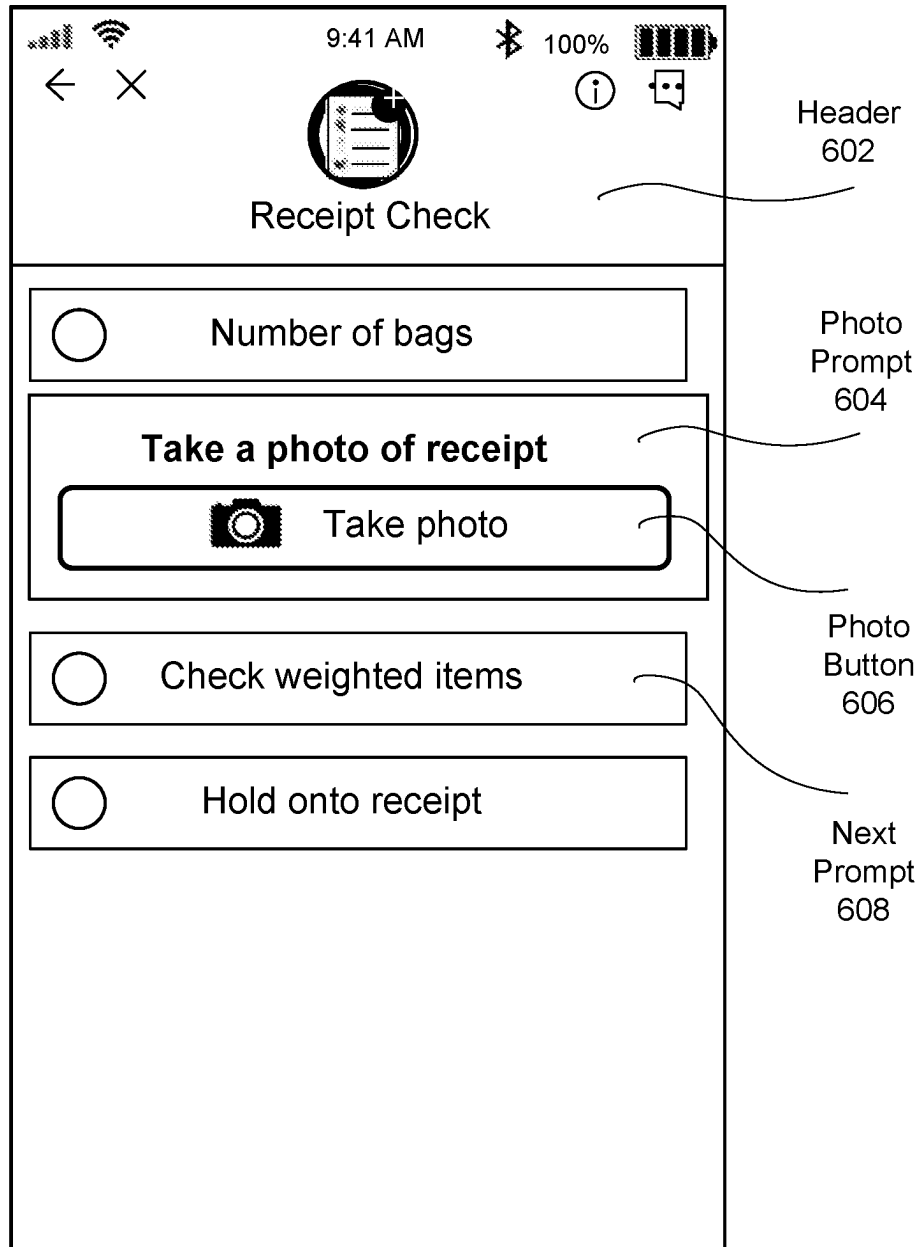


FIG. 6A

610

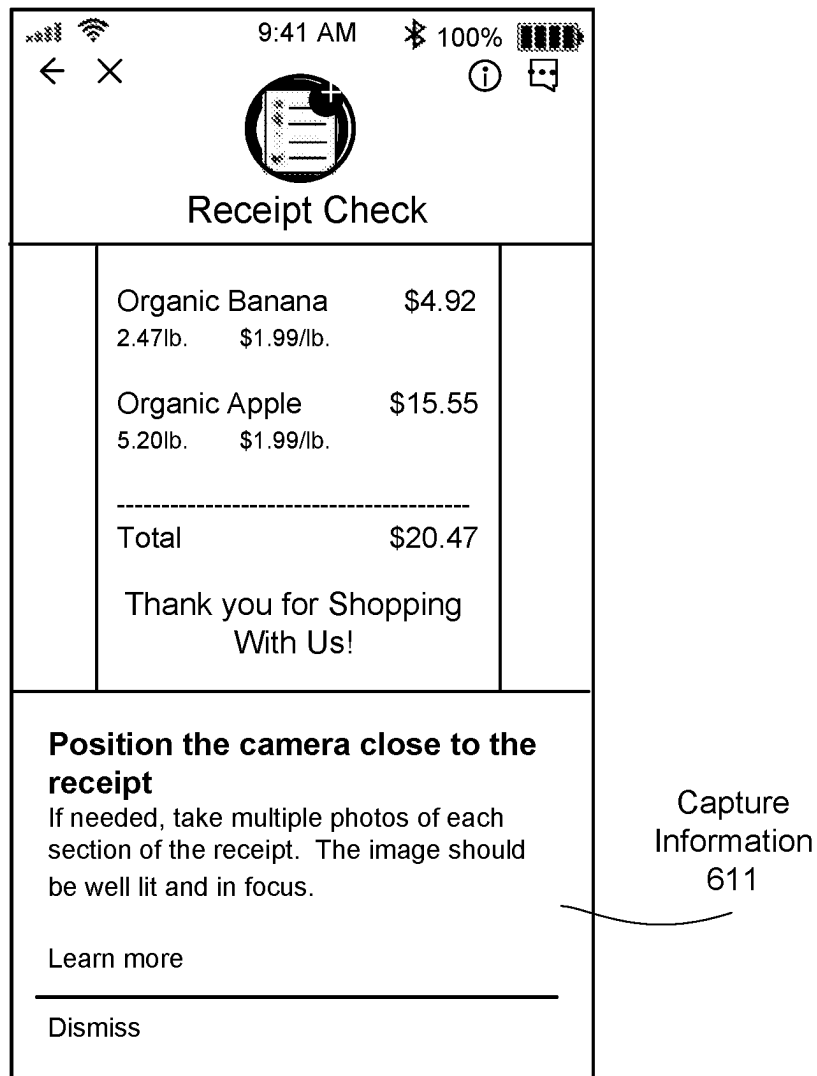
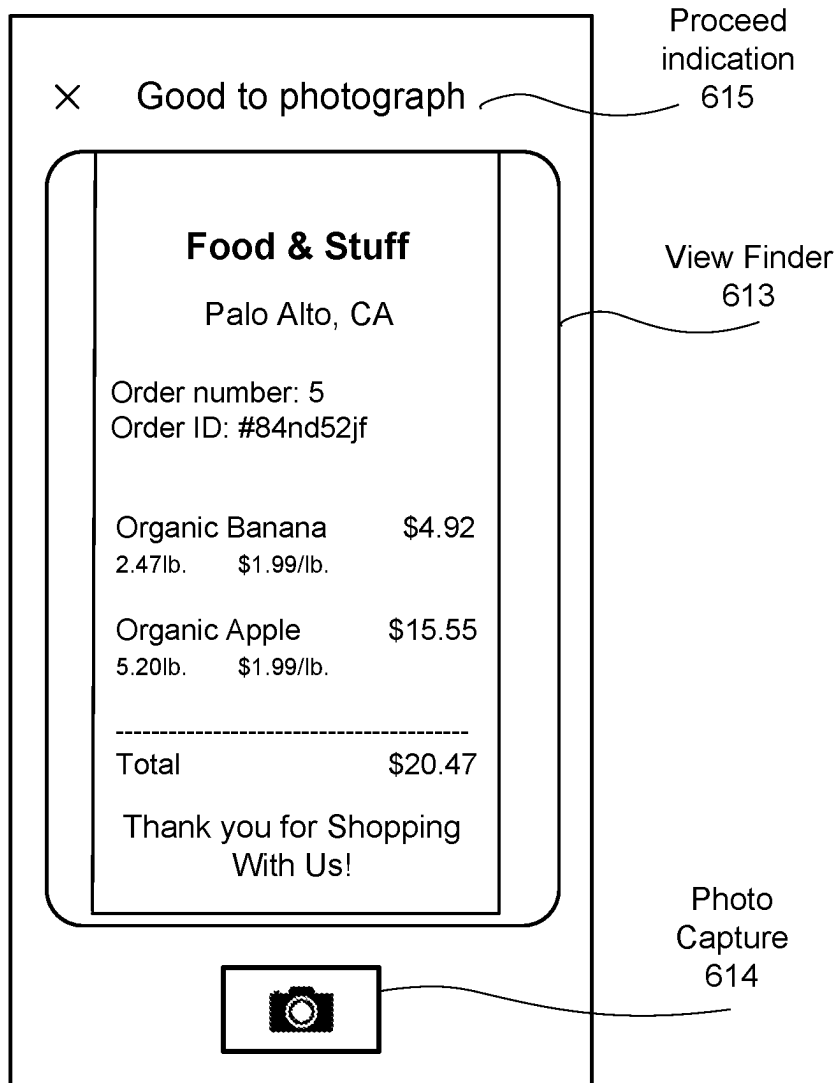


FIG. 6B

612**FIG. 6C**

615

Food & Stuff	
Palo Alto, CA	
Order number: 5 Order ID: #84nd52jf	

Organic Banana	\$4.92
2.47lb. \$1.99/lb.	
Organic Apple	\$15.55
5.20lb. \$1.99/lb.	

1 photo

Retake photo

Add another photo

Submit photo

Cancel

Captured image 616

Interactive elements 617

FIG. 6D

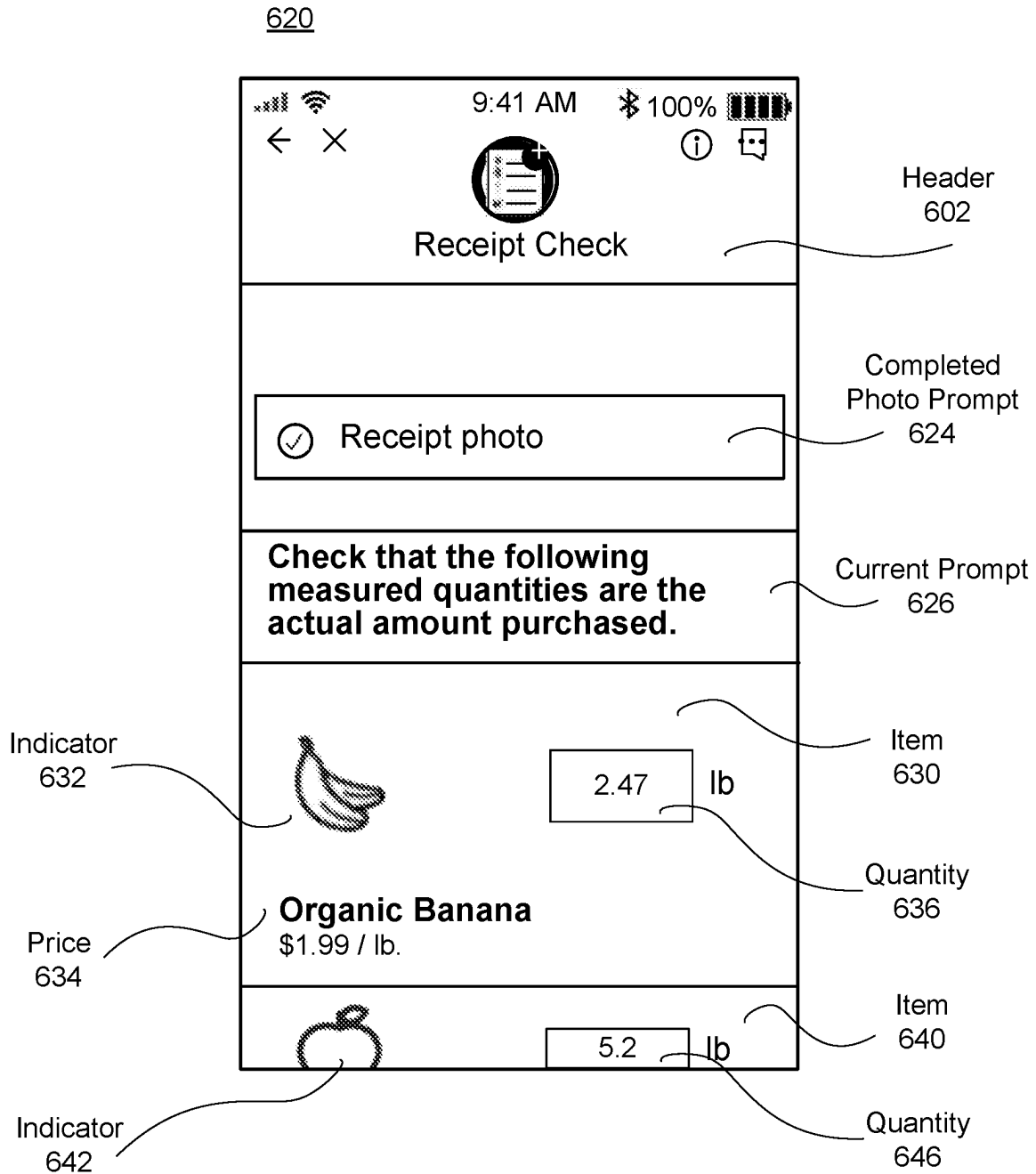


FIG. 6E

650

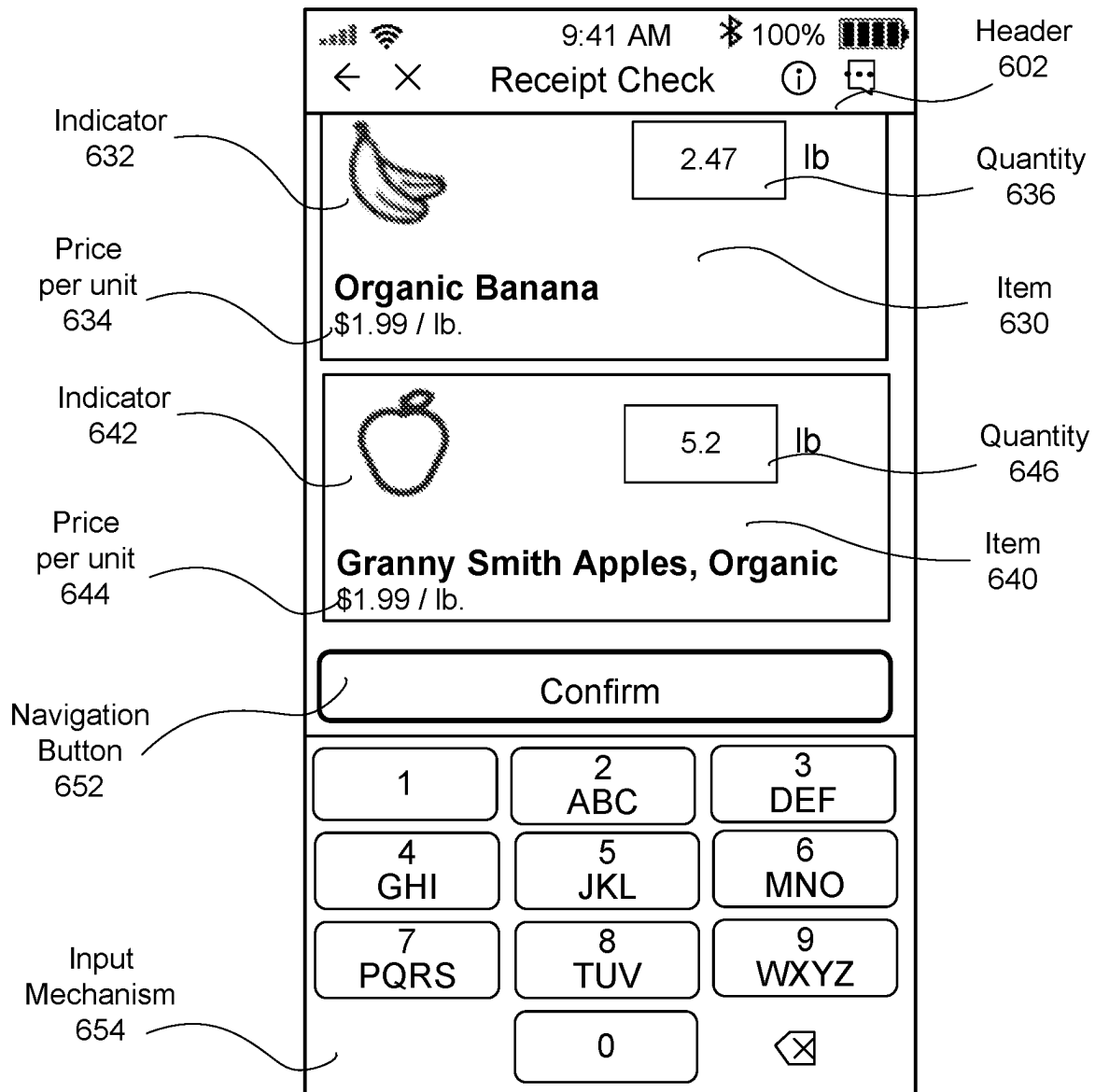


FIG. 6F

700

Image Options 706

Captured Image 702

Zoomed view 704

|| 01:03:24 44 completed 84/hour Shortcuts Insights Log out

Submit button 714

Submit

Good v

Not a receipt v

Not readable v

Incomplete (store issues) v

Fresh Fruits Banana, Organic

2.47 lbs.

Fresh Fruits Apple, Organic

lbs.

Food & Stuff

Palo Alto, CA

Order number: 5

Order ID: #84nd52jf

Organic Banana \$4.92

2.47lb. \$1.99/lb.

Organic Apple \$15.55

5.20lb. \$1.99/lb.

Josh's Bagels \$5.03

5 bagels

Mr. W's Wild Rice \$3.72

1 16 oz. package

FIG. 7

1

OPTICAL SCANNING FOR WEIGHTS AND MEASURES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of co-pending U.S. application Ser. No. 17/068,594, filed Oct. 12, 2020, which is herein incorporated by reference in its entirety.

BACKGROUND

This disclosure relates generally to a process for improving accuracy of a delivery system, and specifically to determining an item quantity based on image analysis of a physical receipt by a computer system.

In current delivery systems, shoppers, or “pickers,” fulfill orders at a physical warehouse, such as a retailer, on behalf of customers as part of an online shopping concierge service. The delivery system provides customers with a user interface that displays an inventory catalog listing items that a customer can add to an order, specifying a quantity of each item. Many items are measured in quantities that are straightforward, such as a dozen eggs, a gallon of milk, or one loaf of bread. Other items may be quantified in weights or other measures that are more difficult to precisely match the quantity specified by the customer. For example, a customer may order 1.0 lb. of apples. At the retailer, the shopper picks 3 apples, which weigh in total 0.96 lbs. An additional apple would bring the total weight significantly over the quantity specified by the customer. However, if the customer pays for the full specified 1.0 lbs. of apples, the customer is being overcharged. In current delivery systems, there is no efficient means for reconciling prices of quantities ordered by a customer and quantities purchased by a shopper for items that have variable weights and measured quantities.

SUMMARY

As described herein, a delivery system generates and uses machine-learned models to identify items and corresponding actual amount purchased in an image of a receipt of an order. In some embodiments, the delivery system identifies items that can be purchased in variable amounts, such as items purchased by weight. The machine-learned models are trained using images of physical receipts, where pickers upload the images of receipts and input known actual amounts purchased. The training data are used to build a deep-learning detection model capable of determining whether the receipt is readable and, if so, to identify items and measured quantities in the receipt representing the actual amounts purchased. The identified measured quantities can be used to charge a customer who placed the order a correct cost for the actual amount purchased.

A method for reconciling cost discrepancies between ordered and purchased amounts of items with variable weights includes receiving an order of items with specified amounts of each item. Each item is associated with a price that is dependent on the actual amount of the item purchased. The method determines an estimated cost of the order based on the price and the ordered amounts of each item. The method sends the order to a shopper for fulfillment at a store. The method receives an image of a receipt for the order from the shopper after fulfillment of the order. The method scans the receipt, using image processing, to identify the items in the receipt. The method identifies a measured

2

quantity associated with each item, the measured quantity representing the actual amount of the item purchased at the store. The method determines an actual price of each item in the order and determines a total cost of the order based on the actual prices.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an environment of an online shopping concierge service, according to one embodiment.

FIG. 2 is a diagram of an online shopping concierge system, according to one embodiment.

FIG. 3A is a diagram of a customer mobile application (CMA), according to one embodiment.

FIG. 3B is a diagram of a picker mobile application (PMA), according to one embodiment.

FIG. 4 is a flowchart illustrating a process for determining an actual price of an item based on discrepancies between an ordered amount and an actual amount purchased of the item, according to one embodiment.

FIG. 5A-5C are example user interfaces for the picker to input training data via the PMA, according to one embodiment.

FIG. 6A-6F are example user interfaces for the picker to upload an image of a receipt via the PMA, according to one embodiment.

FIG. 7 is an example user interface for an auditor to enter measured quantities for a receipt, according to one embodiment.

The figures depict embodiments of the present disclosure for purposes of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles, or benefits touted, of the disclosure described herein.

DETAILED DESCRIPTION

System Overview

FIG. 1 illustrates an environment **100** of an online platform, according to one embodiment. The figures use like reference numerals to identify like elements. A letter after a reference numeral, such as “**110a**,” indicates that the text refers specifically to the element having that particular reference numeral. A reference numeral in the text without a following letter, such as “**110**,” refers to any or all of the elements in the figures bearing that reference numeral. For example, “**110**” in the text refers to reference numerals “**110a**” and/or “**110b**” in the figures.

The environment **100** includes an online concierge system **102**. The online concierge system **102** is configured to receive orders from one or more customers **104** (only one is shown for the sake of simplicity). An order specifies a list of goods (items or products) to be delivered to the customer **104**. The order also specifies the location to which the goods are to be delivered, and a time window during which the goods should be delivered. In some embodiments, the order specifies one or more retailers from which the selected items should be purchased. The customer **104** may use a customer mobile application (CMA) **106** to place the order; the CMA **106** is configured to communicate with the online concierge system **102**.

The online concierge system **102** is configured to transmit orders received from customers **104** to one or more pickers **108**. A picker **108** may be a contractor, employee, or other person (or entity) who is enabled to fulfill orders received by the online concierge system **102**. The picker **108** travels

between a warehouse and a delivery location (e.g., the customer's home or office). A picker **108** may travel by car, truck, bicycle, scooter, foot, or other mode of transportation. In some embodiments, the delivery may be partially or fully automated, e.g., using a self-driving car. The environment **100** also includes three warehouses **110a**, **110b**, and **110c** (only three are shown for the sake of simplicity; the environment could include hundreds of warehouses). The warehouses **110** may be physical retailers, such as grocery stores, discount stores, department stores, etc., or non-public warehouses storing items that can be collected and delivered to customers. Each picker **108** fulfills an order received from the online concierge system **102** at one or more warehouses **110**, delivers the order to the customer **104**, or performs both fulfillment and delivery. In one embodiment, pickers **108** make use of a picker mobile application (PMA) **112** which is configured to interact with the online concierge system **102**.

FIG. 2 is a diagram of an online concierge system **102**, according to one embodiment. The online concierge system **102** includes an inventory management engine **202**, which interacts with inventory systems associated with each warehouse **110**. In one embodiment, the inventory management engine **202** requests and receives inventory information maintained by the warehouse **110**. The inventory of each warehouse **110** is unique and may change over time. The inventory management engine **202** monitors changes in inventory for each participating warehouse **110**. The inventory management engine **202** is also configured to store inventory records in an inventory database **204**. The inventory database **204** may store information in separate records—one for each participating warehouse **110**—or may consolidate or combine inventory information into a unified record. Inventory information includes both quantitative and qualitative information about items, including size, color, weight, SKU, serial number, and so on. The inventory database includes pricing information about each item, such as a unit price, a price per weight, a price per quantity (e.g., the price of a 6-pack of a juice may be more than half of the price of a 12-pack of the same juice), and/or an estimated price thereof (e.g., prices of produce may vary based on season and availability of the produce). The inventory database **204** also stores information about various item properties, such as vegan, organic, gluten free, etc. The inventory database **204** also stores purchasing rules associated with each item, if they exist. For example, age-restricted items such as alcohol and tobacco are flagged accordingly in the inventory database **204**. The inventory management engine **202** may receive updated item information from a picker **104**, such as a property of an item in an inventory database **204**, and adds the updated item information to the inventory database **204**.

The online concierge system **102** includes an order fulfillment engine **206** which is configured to synthesize and display an ordering interface to each customer **104** (for example, via the CMA **106**). The order fulfillment engine **206** is also configured to access an inventory database **204** in order to determine which items are available at which warehouses **110**, and to identify properties associated with the items. The order fulfillment engine **206** determines a sale price for each item ordered by a customer **104**. In one embodiment, the order fulfillment engine determines an estimated price for an order as a whole, based on the sale prices of each item. Prices set by the order fulfillment engine **206** may or may not be identical to in-store prices determined by retailers (which is the price that customers **104** and pickers **108** would pay at retail warehouses). The order

fulfillment engine **206** also facilitates transactions associated with each order. In one embodiment, the order fulfillment engine **206** charges a payment instrument associated with a customer **104** when he/she places an order. The order fulfillment engine **206** may transmit payment information to an external payment gateway or payment processor. The order fulfillment engine **206** stores payment and transactional information associated with each order in a transaction records database **208**.

In some embodiments, the order fulfillment engine **206** also shares order details with warehouses **110**. For example, after successful fulfillment of an order, the order fulfillment engine **206** may transmit a summary of the order to the appropriate warehouses **110**. The summary may indicate the items purchased, the total value of the items, and in some cases, an identity of the picker **108** and customer **104** associated with the transaction. In one embodiment, the order fulfillment engine **206** pushes transaction and/or order details asynchronously to retailer systems. This may be accomplished via use of webhooks, which enable programmatic or system-driven transmission of information between web applications. In another embodiment, retailer systems may be configured to periodically poll the order fulfillment engine **206**, which provides detail of all orders which have been processed since the last request.

The order fulfillment engine **206** may interact with a picker management engine **210**, which manages communication with and utilization of pickers **108**. In one embodiment, the picker management engine **210** receives a new order from the order fulfillment engine **206**. The picker management engine **210** identifies the appropriate warehouse to fulfill the order based on one or more parameters, such as a probability of item availability, the contents of the order, the inventory of the warehouses, and the proximity to the delivery location. The picker management engine **210** then identifies one or more appropriate pickers **108** to fulfill the order based on one or more parameters, such as the pickers' proximity to the appropriate warehouse **110** (and/or to the customer **104**), his/her familiarity level with that particular warehouse **110**, and so on. For example, the picker management engine **210** identifies pickers by comparing the parameters to data retrieved from a picker database **212**. The picker database **210** stores information describing each picker **108**, such as his/her name, gender, rating, previous shopping history, and so on.

As part of fulfilling an order, the order fulfillment engine **206** and/or picker management engine **210** may also access a customer database **214** which stores information describing each customer. This information could include each customer's name, address, gender, shopping preferences, favorite items, stored payment instruments, and so on.

The order fulfillment engine **206** interacts with the image processing module **216** to adjust an estimated cost of an order based on an image of a receipt that contains actual amounts purchased of items. In one embodiment, the order fulfillment engine **206** determines an estimated price of the order based on ordered quantities of items. Upon receiving an image of the receipt for the order, the image processing module **216** determines a price adjustment based on the difference between an ordered amount and an actual amount purchased of each item. Based on the price adjustment, the order fulfillment engine **206** charges the payment instrument associated with the customer **104** an adjusted cost for the order.

In another embodiment, the order fulfillment engine **206** first charges the payment instrument the estimated price of the order. The image processing module **216** determines the

5

price adjustment, and if the actual amount purchased is less than the ordered amount, the order fulfillment engine **206** reimburses the customer **104** the price adjustment. If the actual amount purchased is greater than the ordered amount, the order fulfillment engine **206** charges the payment instrument associated with the customer **104** the price adjustment. Furthermore, the order fulfillment engine **206** may adjust the cost charged for an order based on a net sum of the differences between the ordered amount and actual amount purchased of each item in the order. That is, the order fulfillment engine **206** may determine a price adjustment for each item in the order and charge or reimburse the customer **104** for the order as a whole based on a net sum of the price adjustments.

As part of fulfilling an order, the order fulfillment engine **206** and/or image processing module **216** may also access an order database **230** which stores information describing each order. This information may include a set of items included in an order, a price per unit of each item, a quantity of each item, a total price per item, information about the customer **104** who placed the order, information about the picker **108** who is picking the order, a specified warehouse **110**, or mappings to such information as stored in the inventory database **204**, the transaction records database **208**, the picker database **212**, and/or the customer database **214**. Additionally, the order database **230** may include information about the status of the order such as an order date, a fulfillment date or estimated fulfillment date, a delivery time or estimated delivery time or window, and one or more images of the order receipt. The image of the receipt is processed by the image processing module **216**.

Image Processing

The online concierge system **102** includes an image processing module **216** for processing images of receipts associated with orders. After fulfillment of an order, an image of a receipt of the order is received at the online concierge system **102**, for example from the picker **108** via the PMA **112**. The online concierge system **102** then stores the image in the order database **230**. The image of the receipt is analyzed by the image processing module **216**, which uses one or more image processing algorithms as discussed below to extract text associated with items, weights, and/or prices on the receipt from the image of the receipt. In some embodiments, the image processing module **216** may be partially or wholly implemented by a third-party or a cloud-based model. In some embodiments, the image processing module **216** includes a quality checker **218**, a text identifier **220**, a text extractor **222**, and a text processor **224**. The image processing module **216** may also store a set of training images **226**.

The quality checker **218** determines whether the image is of sufficient quality to resolve the text of the receipt. An image is of sufficient quality, for example, if the image is of a receipt and not blurry. If the quality checker **218** determines the image is of sufficient quality, the image processing continues. If the quality checker **218** determines the image is not of sufficient quality, the image processing module **216** returns via the PMA **112** a prompt to the picker **108** to take another image of the receipt.

The quality checker **218** may be implemented as a machine learning model trained on training images **226** to determine whether the image is of sufficient quality. For example, the quality checker **218** may be trained on training images **226** that include both positive images of receipts (i.e., show a clear and itemized receipt) and negative images of receipts (i.e., are blurry images, images of objects other than receipts). In other embodiments, the quality checker

6

218 may determine the variance of a fast Fourier transform of the image to determine whether the image is blurry.

The text identifier **220** determines locations of text within the image of the receipt. In some embodiments, the text identifier **220** is a machine learning model trained on the training images **226**. The text identifier **220** obtains a bounding box for instances of text in the image of the receipt. In some embodiments, the text identifier **220** and the quality checker **218** may be one machine learning model that returns a null set of bounding boxes if the text cannot be resolved, i.e., the image is not of sufficient quality.

The text extractor **222** determines the words and numerical values of the text contained within the image of the receipt. In some embodiments, the text extractor **222** applies one or more optical character recognition (OCR) algorithms to the bounding boxes determined by the text identifier **220**. In embodiments without the text identifier **220**, the text extractor applies OCR to the whole image to determine the text within the receipt.

The text processor **224** analyzes the text determined by the text extractor **222** to determine items and their associated amount purchased. That is, the text processor **224** identifies text associated with an item description. The text processor **224** further identifies an amount and total price associated with the item. The text processor **224** can identify amounts that are whole values (e.g., 4 bananas) and amounts that are measured quantities (e.g., 0.96 lbs. of apples) associated with variable weight items. The text processor **224** may specifically classify amounts as whole values and measured quantities. The measured quantities representing the actual amount purchased can be compared by the order fulfillment engine **206** to the ordered amount of the item such that the customer **104** is charged appropriately for the variable weight item. In one embodiment, the measured quantities and image of the receipt may be sent for display to one or more auditors via an auditor mobile application, which is connected to the online concierge system **102**. The one or more auditors may approve or reject the measured quantities in view of the receipt. If the auditors reject the measured quantities, they may enter one or more replacement measured quantities.

In some embodiments, the text processor **224** is implemented as a rules-based natural language processing (NLP) algorithm. In other embodiments, the text processor **224** may classify the instances of text into categories, e.g., name of item, amount, total price. Further, the text processor determines the amount and the total price for the item spatially by determining the corresponding text. For example, the numerical values representing amount and total price closest to the item name on the receipt are associated with one another. The text processor **224** identifies the items and their associated amounts and prices for all instances of text within the receipt, as determined by the text extractor **222**.

The training images **226** are a set of images tagged with metadata. The training images **226** are used to train the one or more machine learning models in the image processing module **216**. The training images **226** includes images of receipts from warehouses, and each image is tagged with information, such as bounding boxes and identification of the text printed on the receipt, to train the text identifier **220** and/or the text extractor. The training images **226** may also include blurry images and images of other objects to train the quality checker **218**.

The training images **226** may be tagged based on receipt information manually input to the online concierge system **102** by a picker **108** through the PMA **112**. In some

embodiments, the training images 226 are tagged by the text identifier 220 and text extractor 222 and provided to a picker 108 or an auditor for review. For instance, the picker 108 can either accept the tags as-is or edit the tags via the PMA 112, as discussed in FIGS. 6A-6F. Alternatively, an auditor may accept or edit the tags in an auditor mobile application displaying user interfaces comparable to those shown in FIGS. 6A-6F. In other examples, the training images 226 may be tagged by a third party. The training images 226 may reflect actual conditions of images of receipts in the real world, such as crumpled receipts that may distort labeling or lighting that over or under exposes portions of text.

The quality checker 218, the text identifier 220, and the text extractor 222 are trained by the image processing module 216 on the training images to determine relative weights of kernel functions within each machine learned model to provide a desired output, the outputs as described above in relation to each module. The kernel function weights may be randomly initialized, e.g., from a Gaussian distribution before training. In some embodiments, the image processing module 216 continually trains the quality checker 218, the text identifier 220, and the text extractor 222 responsive to a picker 108 adding new images to the training images 226.

Customer Mobile Application

FIG. 3A is a diagram of the CMA 106, according to one embodiment. The CMA 106 includes an ordering interface 302, which provides an interactive interface with which the customer 104 can browse through and select products and place an order. The CMA 106 also includes a system communication interface 304 which, among other functions, receives inventory information from the online shopping concierge system 102 and transmits order information to the system 102. The system communication interface 304 also enables the customer 104 to submit a complaint or request to view an image of a receipt associated with their order if they believe they have been charged incorrectly for a variable weight item. The CMA 106 also includes a preferences management interface 306 which allows the customer 104 to manage basic information associated with his/her account, such as his/her home address, contact information, and payment instruments. The CMA 106 may communicate the basic information to the online concierge system 102, such that the online concierge system 102 may transmit receipts for the customer's orders including the actual amount of each order. The preferences management interface 306 may also allow the user to manage other details such as his/her favorite or preferred warehouses 110, preferred delivery times, special instructions for delivery, and so on.

Picker Mobile Application

FIG. 3B is a diagram of the PMA 112, according to one embodiment. The PMA 112 includes a barcode scanning module 320 which allows a picker 108 to scan an item at a warehouse 110 (such as a can of soup on the shelf at a grocery store). The barcode scanning module 320 may also include an interface which allows the picker 108 to manually enter information describing an item (such as its serial number, SKU, quantity and/or weight) if a barcode is not available to be scanned. PMA 112 also includes a basket manager 322 which maintains a running record of items collected by the picker 108 for purchase at a warehouse 110. This running record of items is commonly known as a "basket." In one embodiment, the barcode scanning module 320 transmits information describing each item (such as its cost, quantity, weight, etc.) to the basket manager 322, which updates its basket accordingly. The PMA 112 also includes a system communication interface 324 which inter-

acts with the online shopping concierge system 102. For example, the system communication interface 324 receives an order from the system 102 and transmits the contents of a basket of items to the system 102. The PMA 112 also includes an image encoder 326 which encodes the contents of a basket into an image. For example, the image encoder 326 may encode a basket of goods (with an identification of each item) into a QR code which can then be scanned by an employee of the warehouse 110 at check-out.

The PMA 112 includes an imaging module 328, which allows a picker 108 to collect images of receipts via a camera of a mobile device (e.g., cell phone, tablet, or any electronic device with standard communication technologies). In some embodiments, the imaging module 328 additionally allows a picker 108 to collect images of items available at a warehouse when an item in the order is unavailable, e.g., by taking a photograph of one or more items in a warehouse. In another embodiment, the imaging module 328 may also provide an interface for the picker 108 to confirm image of the receipt and/or the identified contents of the receipt, as discussed in greater detail with respect to FIGS. 5A-5C and 6A-6C. In some embodiments, the imaging module 328 also allows a picker 108 or an auditor to indicate the location and type of an item label in a collected image. For example, a picker 108 (or auditor, who may receive the images via an auditor mobile application for review) may draw a bounding box around an instance of text to indicate that the identified pixels are associated with an instance of text, and select a class associated with the instance of text. The imaging module 328 sends the image of the receipt with the tagged instances of text and their associated classes to the online concierge system 102, which are incorporated into the training images 226.

Determining Cost of Order

FIG. 4 is a flowchart illustrating a process 400 for determining an actual price of an item based on discrepancies between an ordered amount and an actual amount purchased of the item, according to one embodiment. The customer 104 creates an order by selecting a set of items for purchase through the ordering interface 302 of the CMA 106. The order may specify the set of items for purchase at one or more warehouses 110 (e.g., retailers). The online concierge system 102 receives 405 the order from the customer 104. The online concierge system 102 may receive 405 the order via the CMA 106, as discussed in related to FIG. 1.

The online concierge system 102 determines 410 an estimated cost associated with the order. The estimated cost is determined 410 based on the quantities of each item specified by the order and an estimated unit price of each item, as stored in the inventory database 204. In one embodiment, the order fulfillment engine 206 may determine 410 the estimated cost. The online concierge system 102 sends the estimated cost for display via the ordering interface 302 of the CMA 106.

The online concierge system 102 sends 420 the order information to the picker 108. The picker 108 receives the order information via the PMA 112. The order fulfillment engine 206 interacts with the picker management engine 210 to select the picker 108 from a set of available pickers.

The picker 108 fulfills 425 the order. The picker 108 goes to one or more warehouses 110 picks and purchases each item as specified in the order. The picker 108 may use the PMA 112 to keep track of the order and the progress of the fulfillment. Subsequent to fulfillment, the picker 108 takes an image of the receipt of the order. In an embodiment where the picker 108 must go to multiple warehouses 110 to fulfill

an order, the picker **108** takes an image of each receipt from each warehouse **110** in the process of fulfilling the order.

The picker **108** sends **430** the image of the receipt to the online concierge system **102**. The picker **108** may take the image of the receipt using the imaging module **328** of the PMA **112**, which integrates with a camera on the picker's mobile device. The image of the receipt can be sent **430** to the online concierge system **102** using the PMA **112**.

The online concierge system **102** performs **435** image processing on the image of the receipt. The image processing is performed by the image processing module **216**. As described in relation to the image processing module **216**, the image processing includes machine learning models and OCR to identify and extract text from the image of the receipt. In particular, the text identifier **220** determines location of text within the receipt, such as by a bounding box, and the text extractor **222** determines the words and numerical values of the text, for example using OCR. Further, in some embodiments, the image processing may contain an initial step, performed by the quality checker **218**, of determining whether the image is of sufficient quality to resolve the text of the receipt.

The online concierge system **102** identifies **440** a measured quantity within the image of the receipt. Based on the words and numerical values determined by the image processing, the online concierge system **102** uses the text processor **224** to determine an item on the receipt and its associated amount, such as a measured quantity, and its associated price. The amount and price are reflective of that actually purchased by the picker since the values are extracted from the image of the receipt.

The online concierge system **102** determines **445** an actual price of the item in the order based on the measured quantity identified within the image. In one embodiment, the order fulfillment engine **206** determines an actual price by calculating the actual price based on the measured quantity the item and an estimated unit price of the item. Furthermore, the online concierge system **102** may determine an actual price of multiple items in the order if multiple items are associated with measured quantities.

The online concierge system **102** determines **460** a total cost of the order based on the actual price of the item (or items, in some embodiments). In particular, the online concierge system **102** sums the prices of each item in the order using the actual price of items determined based on measured quantities. The online concierge system **102** charges **465** the customer for the total cost of the order and may send the total cost for display via the ordering interface **302** of the CMA **106**.

In some embodiments, the online concierge system **102** may charge the customer **104** for the estimated cost of the order, and subsequently determines a price adjustment for the order based on the difference between the quantity specified in the order by the customer **104** and the identified **440** quantity of the item in the receipt. If the quantity of the item identified **440** in the receipt is the same as that specified by the customer **104** in the order, the adjustment may be null. In other embodiments, the order fulfillment engine **206** determines a price adjustment based on the difference between the estimated price of an item provided to the customer **104** and the price for the item extracted from the image of the receipt. Furthermore, in these embodiments, the order fulfillment engine may adjust the cost of the order based on a net sum of adjustments for each item in the order.

In these embodiments, the online concierge system **102** sends an adjusted price to the customer **104** based on the actual amounts purchased in the order where the adjusted

price is the actual cost of the order. The order fulfillment engine **206** either charges or reimburses the customer **104** based on the price adjustment, or the net price adjustment across all items in the order. The order fulfillment engine **206** accesses the transaction records database **208** to appropriately charge or refund the payment instrument of the customer **104**.

The process **400** provides an improved method for delivery systems. The customer **104** is charged the correct amount based on the quantity of items actually purchased. Therefore, the customer **104** is not overcharged for the items in the order. Additionally, the automated image processing and analysis performed by the online concierge system **102** to determine the amounts actually purchased of each item in the receipt based on an image reduces the input time and effort of the picker **108**. The process **400** improves equitability of cost.

FIG. 5A-5C are example user interfaces for the picker to input training data via the PMA, according to one embodiment. The user interfaces shown in FIGS. 5A-5C are example embodiments of displays generated by the PMA **112** on a mobile device. Alternate embodiments may include additional or fewer features or otherwise be structured and displayed differently in a user interface. The data input via the example user interfaces of FIGS. 5A-5C is stored in the training images **226**.

FIG. 5A shows a user interface **500** prompting the picker **108** to capture an image of the receipt. The interface **500** may be displayed to the picker **108** via the PMA **112**, in particular via the imaging module **328**, subsequent to purchase of some or all of a set of items specified in an order. For example, the user interface **500** may be provided for display by the imaging module **328** responsive to the PMA **112** receiving an indication from the picker **108** that one or more items in the order have been purchased.

The user interface **500** includes a header **502**, which includes an indication that this stage of the fulfillment of the order is a receipt check. The header **502** also includes other indications of time, data signal, and/or battery, as consistent with the operating system of the mobile device running the PMA **112**. The user interface **500** includes a photo prompt **504** which indicates an action item for the picker **108** is to take a photo of the receipt. The photo prompt **504** includes a photo button **506**, which the picker **108** selects to take a photo of the receipt. When selected, the photo button launches a camera attached to or associated with the mobile device running the PMA **112**. The picker **108** uses the camera to take an image of the receipt. In another embodiment, the photo prompt **504** enables the picker to upload an existing image of the receipt. The user interface **500** also includes a next prompt **508**, which indicates a future action item is to check weighted items, as discussed in greater detail below.

FIG. 5B shows a user interface **510** prompting the picker **108** to enter an actual amount purchased of each item on the receipt. The user interface **510** includes the header **502**. The user interface **510** also includes a completed photo prompt **514** indicating that the picker **108** has successfully uploaded a photo using the imaging module **328**.

The user interface **510** includes a current prompt **516** prompting the picker **108** to enter the actual amount purchased of each item as seen on the receipt. The interface **510** further includes a first item **520** for which the picker has been prompted to enter the actual amount purchased. The first item **520** includes an indicator **522**, which may be an image, a graphic, and/or a name of the first item **520**. The first item **520** includes a price per unit **524** that indicates a

11

unit price or price per weight of the first item **520**. The first item **520** includes a quantity **526**, which is an empty field for the user to enter the actual amount purchased of the first item **520** as printed on the receipt.

In one embodiment, the PMA **112** provides for display the first item **520**, including the indicator **522** and the price per unit **524**, based on data received about the order from the online concierge system **102**. In particular, the indicator **522** and the price per unit **524** may be data about the first item **520** that has been stored in the inventory database **204**. That is, the PMA **112** provides for display each of the items in the order that have a variable weight for the picker **108** to provide the actual amount purchased. In some embodiments, the PMA **112** provides for display all the items included in the order (i.e., not just the variable weight items) to be able to tag and classify each instance of text in the image of the receipt for use in the training images **226**. The picker **108** selects the quantity **526** and manually inputs the actual amount purchased on the first item **520** as printed on the receipt.

In some embodiments, the user interface **510** includes a second item **530** for which the picker has been prompted to enter the actual amount purchased. The PMA **112** may provide for display fields including an indicator, a price per unit, and a quantity for the item **530** as the picker **108** interacts with the user interface **510** (e.g., completes the quantity **526** for the first item **520**, scrolls down the user interface **510**).

FIG. **5C** shows a user interface **540** prompting the picker **108** to enter an actual amount purchased of the second item **530** on the receipt. The user interface **540** may be provided for display by the PMA **112** after the picker **108** inputs the actual amount purchased of the first item **520**. The user interface **540** includes the header **502**, which is reduced in size as the user navigates through the items **520**, **530**. The user interface includes the item **520**, further including the indicator **522**, the price per unit **524**, and the quantity **526**. In the embodiment shown by the user interface **540**, the picker has manually input the actual amount purchased of 2.3 lbs. in the field for the quantity **526**.

The user interface **540** includes the second item **530** for which the picker has been prompted to enter the actual amount purchased. The second item **530** includes an indicator **532**, which may be an image, a graphic, and/or a name of the second item **530**. The second item **530** includes a price per unit **534** that indicates a unit price or price per weight of the second item **530**. The second item **530** includes a quantity **536**, which is an empty field for the user to enter the actual amount purchased of the second item **530** as printed on the receipt. In one embodiment, the PMA **112** provides for display the second item **530**, including the indicator **532** and the price per unit **534**, based on data received about the order from the online concierge system **102**. In particular, the indicator **532** and the price per unit **534** may be data about the second item **530** that has been stored in the inventory database **204**.

The user interface **540** further includes a navigation button **542**, which when selected by the picker **108** navigates to the next item in the order. After the picker **108** has manually input the actual amount purchased in the quantity **536**, the picker can select the navigation button **542** to manually input the actual amount purchased of the next item. The user interface **540** also includes an input mechanism **544** which enables the picker **108** to manually input values into the quantity **526**, **536** fields. The input mechanism **544** may be a keyboard-like user interface provided for display and interaction as shown in the embodiment of FIG.

12

5C. In other embodiments, an input mechanism may be a physical keyboard, speech-to-text device, or any other means of manually inputting values.

The data input by the picker **108** via the user interfaces **500**, **510**, **540**, including the image of the receipt and actual amounts purchased of items, is stored in the training images **226** and used to train machine learned models, such as the quality checker **218**, the text identifier **220**, and the text processor **224**.

FIG. **6A-6F** are example user interfaces for the picker to upload an image of a receipt via the PMA, according to one embodiment. The user interfaces shown in FIGS. **6A-6F** are example embodiments of displays generated by the PMA **112** on a mobile device. Alternative embodiments may include additional or fewer features or otherwise be structured and displayed differently in a user interface. The data input via the example user interfaces of FIGS. **6A-6F** is stored in the order database **230**. The example interfaces shown in FIGS. **6A-6F** are used to properly charge the customer **104** for actual amounts purchased of items in the order.

FIG. **6A** shows a user interface **600** prompting the picker **108** to capture an image of the receipt. The interface **600** may be displayed to the picker **108** via the PMA **112**, in particular via the imaging module **328**, subsequent to purchase of some or all of a set of items specified in an order. For example, the user interface **600** may be provided for display by the imaging module **328** responsive to the PMA **112** receiving an indication from the picker **108** that one or more items in the order have been purchased.

The user interface **600** includes a header **602**, which includes an indication that this stage of the fulfillment of the order is a receipt check. The header **602** also includes other indications of time, data signal, and/or battery, as consistent with the operating system of the mobile device running the PMA **112**. The user interface **600** includes a photo prompt **604** which indicates an action item for the picker **108** is to take a photo of the receipt. The photo prompt **604** includes a photo button **606**, which the picker **108** selects to take a photo of the receipt. When selected, the photo button launches a camera attached to or associated with the mobile device running the PMA **112**. The picker **108** uses the camera to take an image of the receipt. In another embodiment, the photo prompt **604** enables the picker to upload an existing image of the receipt. The user interface **600** also includes a next prompt **608**, which indicates a future action item is to check weighted items, as discussed in greater detail below, and may include more prompts before or after the photo prompt **604** and next prompt **608**. In some instances, the user interface **600** may additionally include a text box that shares feedback with the picker **108** on the quality of the image and/or positioning recommendations for the camera to improve image quality as the picker **108** uses the camera to take images.

FIG. **6B** shows a user interface **610** including capture information **611** for the picker capturing the image. The capture information includes a description of how to position the camera to capture the picture and describes that in some instances, the picker may need to capture multiple images to capture the entire receipt. The user interface **610** may include further widgets the user may interact with to dismiss the capture information **611**, learn more about the capture information, and exit out of the user interface **610**.

FIG. **6C** shows a user interface **612** enabling the picker **108** to capture an image of the receipt. The user interface **612** shows a view finder **613** enabling the picker **108** to view the area the imaging module **328** is ready to capture. In the

13

embodiment shown in FIG. 6C, the view finder 613 contains an image of the receipt 614. The user interface 612 includes a proceed indication 615 that indicates whether the image of the receipt as shown in the view finder 613 is ready for capture based on an assessment of image quality. To capture the image of the receipt 614 shown in the view finder 613, the picker selects photo capture 614. When the picker 108 selects the photo capture 614, the image of the receipt in the view finder 613 is captured and provided by the PMA 112 to the online concierge system 102. In some embodiments, if the receipt is double-sided or too long to fit in one image, the picker may capture multiple images that are provided to the PMA 112. The image processing module 216 processes the image of the receipt and identifies the actual amount purchased of each item.

FIG. 6D shows a user interface 615 depicting the captured image 616 of the receipt. The user interface 615 prompts the picker 108 to select one of the interactive elements 617 to either retake the captured image, which leads back to the user interface 612 shown in FIG. 6C, add another image of the receipt, submit the captured image 616, or cancel the uploading the captured image 616 of the receipt. If the picker 108 selects to submit the photo via the interactive elements 617, the PMA 112 may display the user interface of FIG. 6E. Alternatively, the online concierge system 102 may transmit the captured image 616 (or images, in some embodiments) to an auditor mobile application, as shown in FIG. 7.

FIG. 6E shows a user interface 620 prompting the picker 108 to check each measured quantity of each item on the receipt is the actual amount purchased. The user interface 620 includes the header 602. The user interface 620 also includes a completed photo prompt 624 indicating that the picker 108 has successfully uploaded a photo using the imaging module 328.

The user interface 620 includes a current prompt 626 prompting the picker 108 to check that each identified measured quantity is the actual amount purchased of each item. That is, the picker 108 is checking that the image processing module 216 correctly identified the actual amount purchased of each item. The user interface 620 further includes a first item 630 for which the picker has been prompted to check that the measured quantity is the actual amount purchased. The first item 630 includes an indicator 632, which may be an image, a graphic, and/or a name of the first item 630. The first item 630 includes a price per unit 634 that indicates a unit price or price per weight of the first item 630. The first item 630 includes a quantity 636, which is a field populated with a measured quantity of the first item 630 as identified in the image of the receipt by the image processing module 216.

In one embodiment, the PMA 112 provides for display the first item 630, including the indicator 632 and the price per unit 634, based on data received about the order from the online concierge system 102. In particular, the indicator 632 and the price per unit 634 may be data about the first item 630 that has been stored in the inventory database 204. In another embodiment, the image processing module 216 identifies the items, prices, and quantities from the image of the receipt and provides them for display to the PMA 112. The PMA 112 provides for display each of the variable weight items in the order for the picker 108 to check the actual amount purchased is correct.

In some embodiments, the user interface 620 includes a second item 640, which also includes an indicator 642, a price per unit 644 (shown in FIG. 6F), and a quantity 646. The PMA 112 may provide for display the indicator 642, the

14

price per unit 644, and the quantity 646 in full as the picker 108 interacts with the user interface 620 (e.g., checks the quantity 636 for the first item 630, scrolls down the user interface 620).

FIG. 6F shows a user interface 650 prompting the picker 108 to check whether the measured quantity of the second item 640 is the actual amount purchased of the second item 640. The user interface 650 may be provided for display by the PMA 112 after the picker 108 verifies the quantity 636 is the actual amount purchased of the first item 630. The user interface 650 includes the header 602, which is reduced in size as the user navigates through the items 630, 640. The user interface includes the item 630, further including the indicator 632, the price per unit 634, and the quantity 636. In the embodiment shown by the user interface 650, the picker has manually confirmed the actual amount purchased of 2.47 lbs. in the field for the quantity 636.

The user interface 650 includes the second item 640 for which the picker has been prompted to confirm whether the quantity 646 is the actual amount purchased. The quantity 646 field has been populated based on the measured quantity of the second item 640 identified by the image processing module. The picker 108 uses a navigation button 652 to manually confirm the quantity 646 matches the actual amount purchased of the second item 640 as printed on the receipt. Selecting the navigation button 652 the confirm the actual amount purchased also navigates the picker to the next item to be confirmed.

If the quantity 646 identified by the image processing module 216 is incorrect (i.e., does not match the actual amount purchased as printed on the receipt), the picker 108 uses the input mechanism 654 to manually input the actual amount purchased of the second item 640. The input mechanism 654 is substantially similar to the input mechanism 544. In some embodiments, when the input mechanism 654 is used to correct the quantity 646 with the actual amount purchased, the associated data is used to re-train the machine-learned models. That is, the actual amount purchased and the image of the receipt are added to the training images 226 and used by the image processing module 216 to re-train the quality checker 218, the text identifier 220, and the text extractor 222.

When the picker 108 has finished verifying that all measured quantities are the actual amounts purchased, or otherwise inputting the actual amounts purchased, the PMA 112 provides the actual amounts purchased to the online concierge system 102. The online concierge system 102 charges the customer 104 based on the amounts actually purchased, as previously discussed.

FIG. 7 is an example user interface 700 for an auditor to enter measured quantities for a receipt, according to one embodiment. The user interface shown in FIG. 7 is an example embodiment of displays generated by an auditor mobile application 112 on a computing device. The computing device may be a mobile device or any suitable computing device. Alternative embodiments of FIG. 7 may include additional or fewer features or otherwise be structured and displayed differently in a user interface. The data input via the example user interface of FIG. 7 is stored in the order database 230. The example user interface 700 shown in FIG. 7 is used to determine an actual amount of an order and properly charge the customer 104 the actual amount instead of an estimated cost for an order.

The user interface 700 allows an auditor to check that each identified measured quantity is the actual amount purchased of each item. That is, the auditor checking that the image processing module 216 correctly identified the actual

15

amount purchased of each item. The user interface 700 may be displayed to the auditor subsequent to purchase of some or all of a set of items specified in an order. For example, the user interface 700 may be provided for display to the auditor by the imaging module 328 responsive to the PMA 112 receiving an indication from the picker 108 that one or more items in the order have been purchased.

The user interface 700 includes a captured image 702 of a receipt of the order. In embodiment where the receipt is pictured in multiple images, the auditor may choose which captured image 702 of the receipt to view via the image options 702. The user interface 700 also includes a zoomed view 704 of a portion of the receipt, which the auditor may move around via the user interface 700 to view magnified portions of the captured image 702. The user interface may display statistics for the auditor, such as time spent reviewing receipts, and number of receipts reviewed, number of receipts viewed over time and include widgets for viewing shortcuts, viewing insights, or logging out of the auditor's auditing account.

The user interface 700 includes a number of interactive elements 708 that the auditor may interact with to indicate information about the captured image 702. For instance, the auditor may interact with the interactive elements 708 of the user interface 700 to indicate that the captured image 702 of the receipt is good for use, is not actually a receipt, is unreadable, or is incomplete, among other information. The user interface 700 also includes text fields (or other interactive elements) in which the auditor may enter an actual quantity 712 of an item 710 purchased for the order based on the captured image 702. In some embodiments, the user interface 700 may additionally display an amount of each item 710 identified by image processing module 216. The text fields correspond to items 710 shown in the zoomed view 704 and may update as the auditor moves the zoomed view 704. In the embodiment shown in FIG. 7, the auditor has manually input the actual amount of bananas purchased of 2.47 lbs. in the text field for the quantity 526.

Furthermore, the user interface 700 includes a submit button 714, which the auditor may interact with to confirm the information entered via the interactive elements 708 and quantities 712 entered for the items 710. The auditor may interact with the submit button 714 to indicate that all quantities 712 on the receipt have been entered. When the auditor 108 has finished entering all quantities 712, the auditor mobile application provides the actual amounts (e.g., quantities 712) purchased to the online concierge system 102. The online concierge system 102 may access the inventory database 204 to determine a correct cost (i.e., actual cost) of each item and charges the customer 104 based on the amounts actually purchased, as previously discussed.

In some embodiments, each captured image 702 of a receipt may be shown via user interface 700 to an auditor for review. In other embodiments, an auditor is only shown a captured image of a receipt if one or more quantities 712 identified by the image processing module 216 are incorrect (i.e., does not match the actual amount purchased as printed on the receipt). In these embodiments, the auditor manually input the quantity 712 of each item purchased. Furthermore, in some instances, when the auditor corrects one or more quantities 712 for a receipt, the associated data is used to re-train the machine-learned models described above. That is, the actual amount purchased and the captured image 702 of the receipt are added to the training images 226 and used by the image processing module 216 to re-train the quality checker 218, the text identifier 220, and the text extractor 222. The foregoing description of the embodiments of the

16

invention has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

Some portions of this description describe the embodiments of the invention in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, micro-code, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In one embodiment, a software module is implemented with a computer program product comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

Embodiments of the invention may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a tangible computer readable storage medium, which includes any type of tangible media suitable for storing electronic instructions, and coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

Embodiments of the invention may also relate to a computer data signal embodied in a carrier wave, where the computer data signal includes any embodiment of a computer program product or other data combination described herein. The computer data signal is a product that is presented in a tangible medium or carrier wave and modulated or otherwise encoded in the carrier wave, which is tangible, and transmitted according to any suitable transmission method.

Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the invention be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments of the invention is intended to be illustrative, but not limiting, of the scope of the invention, which is set forth in the following claims.

What is claimed is:

1. A method implemented at an online system including one or more processors, the method comprising:
 - 5 sending an order associated with a customer to a shopper for fulfillment at a store, the order specifying an ordered amount of an item;

17

receiving an image of a receipt for the order from the shopper after fulfillment of the order;

applying, by the one or more processors, one or more machine-learning models to the image of the receipt to identify the item in the image of the receipt, wherein the one or more machine-learning models include:

- a first machine-learning model trained to determine one or more bounding boxes for instances of text in the image of the receipt; and
- a second machine-learning model trained to determine text within the determined one or more bounding boxes,

wherein the first machine-learning model and the second machine-learning model are trained on historical receipt images labeled with bounding boxes and identification of text within the bounding boxes;

identifying, by the one or more processors, a measured quantity within the image, the measured quantity representing an actual amount of the item purchased at the store;

receiving, by the one or more processors, either a confirmation from the shopper that the identified measured quantity within the image is correct, or a correction to the identified measured quantity provided by the shopper as an annotation, the first machine-learning model and the second machine-learning model are retrained using the image of the receipt and the annotation, thereby improving accuracy of identifying measured quantities in subsequent receipt images;

determining, by the one or more processors, a difference between the actual amount and the ordered amount of the item;

determining, by the one or more processors, an amount to be charged or reimbursed to the customer based in part on the difference between the actual amount and the ordered amount of the item;

determining an estimated cost for the order based on the ordered amount;

charging the customer for the order based on an estimated cost;

responsive to the actual amount being greater than the ordered amount, charging the customer a price adjustment based on the actual price; and

responsive to the actual amount being less than the ordered amount, reimbursing the customer the price adjustment based on the actual price.

2. The method of claim 1, further comprising:

displaying the image of the receipt to an auditor;

displaying, to the auditor, a notification comprising a selectable option to confirm whether the image is readable.

3. The method of claim 1, further comprising:

displaying, to an auditor, a notification comprising the identified item, the identified measured quantity, an option to confirm the identified measured quantity is the ordered amount, and an alternate option to decline the identified measured quantity and input a correct ordered amount.

4. The method of claim 3, further comprising:

displaying, to the auditor, a highlighted version of the image, wherein the identified item and the identified measured quantity are highlighted.

5. The method of claim 1, further comprising:

determining whether the image of the receipt is of sufficient quality to resolve text of the receipt;

18

responsive to determining the image of the receipt is not of sufficient quality, prompting the shopper to take another image of the receipt.

6. The method of claim 1, wherein identifying the measured quantity within the image comprises:

- displaying, to an auditor, an image of the receipt; and
- receiving, from the auditor, the measured quantity of the item entered via a user interface presented to the auditor.

7. The method of claim 1, further comprising:

- determining a total cost of the order based on the actual amount of the item; and
- charging the customer based on the total cost.

8. A non-transitory computer-readable storage medium storing instructions, the instructions when executed cause one or more processors to:

- send an order associated with a customer to a shopper for fulfillment at a store, the order specifying an ordered amount of an item;
- receive an image of a receipt for the order from the shopper after fulfillment of the order;
- apply one or more machine-learning models to the image of the receipt to identify the item in the image of the receipt, wherein the one or more machine-learning models include:
 - a first machine-learning model trained to determine one or more bounding boxes for instances of text in the image of the receipt; and
 - a second machine-learning model trained to determine text within the determined one or more bounding boxes,
- wherein the first machine-learning model and the second machine-learning model are trained on historical receipt images labeled with bounding boxes and identification of text within the bounding boxes;
- identify a measured quantity within the image, the measured quantity representing an actual amount of the item purchased at the store;
- receive, by the one or more processors, either a confirmation from the shopper that the identified measured quantity within the image is correct, or a correction to the identified measured quantity provided by the shopper as an annotation, the first machine-learning model and the second machine-learning model are retrained using the image of the receipt and the annotation, thereby improving accuracy of identifying measured quantities in subsequent receipt images;
- determine a difference between the actual amount and the ordered amount of the item;
- determine an amount to be charged or reimbursed to the customer based in part on the difference between the actual amount and the ordered amount of the item;
- determine an estimated cost for the order based on the ordered amount;
- charge the customer for the order based on an estimated cost;
- responsive to the actual amount being greater than the ordered amount, charge the customer a price adjustment based on the actual price; and
- responsive to the actual amount being less than the ordered amount, reimburse the customer the price adjustment based on the actual price.

9. The non-transitory computer-readable storage medium of claim 8, storing additional instructions, the additional instructions when executed cause the one or more processors to:

- display the image of the receipt to an auditor;

19

display, to the auditor, a notification comprising a selectable option to confirm whether the image is readable.

10. The non-transitory computer-readable storage medium of claim 8, storing additional instructions, the additional instructions when executed cause the one or more processors to:

display, to an auditor, a notification comprising the identified item, the identified measured quantity, an option to confirm the identified measured quantity is the ordered amount, and an alternate option to decline the identified measured quantity and input a correct ordered amount.

11. The non-transitory computer-readable storage medium of claim 10, storing additional instructions, the additional instructions when executed cause the one or more processors to:

display, to the auditor, a highlighted version of the image, wherein the identified item and the identified measured quantity are highlighted.

12. The non-transitory computer-readable storage medium of claim 8, storing additional instructions, the additional instructions when executed cause the one or more processors to:

determine whether the image of the receipt is of sufficient quality to resolve text of the receipt; and responsive to determining the image of the receipt is not of sufficient quality, prompt the shopper to take another image of the receipt.

13. The non-transitory computer-readable storage medium of claim 8, wherein identifying the measured quantity within the image comprises:

displaying, to an auditor, an image of the receipt; and receiving, from the auditor, the measured quantity of the item entered via a user interface presented to the auditor.

14. The non-transitory computer-readable storage medium of claim 8, storing additional instructions, the additional instructions when executed cause the one or more processors to:

determine a total cost of the order based on the actual amount of the item; and charge the customer based on the total cost.

15. A computer system, comprising one or more processors, and a non-transitory computer-readable storage medium storing instructions, the instructions when executed cause the one or more processors to:

send an order associated with a customer to a shopper for fulfillment at a store, the order specifying an ordered amount of an item;

receive an image of a receipt for the order from the shopper after fulfillment of the order;

apply one or more machine-learning models to the image of the receipt to identify the item in the image of the receipt, wherein the one or more machine-learning models include:

a first machine-learning model trained to determine one or more bounding boxes for instances of text in the image of the receipt; and

a second machine-learning model trained to determine text within the determined one or more bounding boxes,

wherein the first machine-learning model and the second machine-learning model are trained on historical

20

receipt images labeled with bounding boxes and identification of text within the bounding boxes;

identify a measured quantity within the image, the measured quantity representing an actual amount of the item purchased at the store;

receive, by the one or more processors, either a confirmation from the shopper that the identified measured quantity within the image is correct, or a correction to the identified measured quantity provided by the shopper as an annotation, the first machine-learning model and the second machine-learning model are retrained using the image of the receipt and the annotation, thereby improving accuracy of identifying measured quantities in subsequent receipt images;

determine a difference between the actual amount and the ordered amount of the item;

determine an amount to be charged or reimbursed to the customer based in part on the difference between the actual amount and the ordered amount of the item;

determine an estimated cost for the order based on the ordered amount;

charge the customer for the order based on an estimated cost;

responsive to the actual amount being greater than the ordered amount, charge the customer a price adjustment based on the actual price; and

responsive to the actual amount being less than the ordered amount, reimburse the customer the price adjustment based on the actual price.

16. The system of claim 15, wherein the one or more processors are further caused to:

display the image of the receipt to an auditor;

display, to the auditor, a notification comprising a selectable option to confirm whether the image is readable.

17. The system of claim 15, wherein the one or more processors are further caused to:

display, to an auditor, a notification comprising the identified item, the identified measured quantity, an option to confirm the identified measured quantity is the ordered amount, and an alternate option to decline the identified measured quantity and input a correct ordered amount.

18. The system of claim 17, wherein the one or more processors are further caused to:

display, to the auditor, a highlighted version of the image, wherein the identified item and the identified measured quantity are highlighted.

19. The system of claim 15, wherein the one or more processors are further caused to:

determine whether the image of the receipt is of sufficient quality to resolve text of the receipt;

responsive to determining the image of the receipt is not of sufficient quality, prompt the shopper to take another image of the receipt.

20. The system of claim 15, wherein identifying the measured quantity within the image comprises:

displaying, to an auditor, an image of the receipt; and

receiving, from the auditor, the measured quantity of the item entered via a user interface presented to the auditor.

* * * * *